

Lecture 9

From STL Mesh to Basic Blender Rendered Animation

Engineering and Computer Graphics
Module: Computer Graphic with Blender
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Minimal functional project for today:

import dragon.stl, build a simple scene, animate a flight path, and render a short shot.

STL → scene → material → world lighting → path animation → final render

Why this lecture matters

Project brief

Today's minimal functional project

- Input asset: one provided dragon STL mesh.
- Build a readable scene instead of a complex production pipeline.
- Animate the dragon as a flying object using a curve-based path.
- Render one still frame first, then render the short motion sequence.

Teaching method: demo one action, students repeat immediately, then instructor checks the result before moving on.

What the class is really practicing

1. Translate a CAD-style mesh into a renderable Blender scene.
2. Connect materials, world lighting, camera, and animation in one workflow.
3. Make decisions that are “good enough” for classroom rendering speed.
4. Separate preview settings from final output settings.

Learning outcomes and visible deliverables

Outcomes

By the end of lecture 9, each student should be able to:

1. Import an STL file into Blender and keep orientation / scale under control.
2. Create a basic renderable setup with material, world lighting, and camera.
3. Animate the dragon moving through space with a controlled path.
4. Render a short animation using frame sequence workflow and export a video.

Lecture evidence of success

- Viewport shows the imported dragon correctly.
- A still render is readable and intentionally lit.
- Timeline plays a full flight from start to end frame.
- Output folder contains numbered frames and one encoded video.

Classroom rule: do not chase perfect realism first. Confirm visibility, motion, and output correctness before beautification.

FreeCAD → Blender: the mental model shift

Concept

What students already know from FreeCAD

- Curves define shape or motion references.
- Bodies have scale, orientation, and local coordinates.
- A model can be exported as a mesh for downstream use.



What Blender adds for today's task

- Scene graph: objects, camera, lights, and world all matter.
- Material inputs behave like render-shader controls students already know from OpenGL ideas.

Carry over your geometry discipline; do not carry over the expectation of a feature tree driving everything.

Today's translation: CAD mesh becomes one scene object that we illuminate, frame, animate, and output.

The Blender workspace we will actually touch today

Workspace

1. 3D Viewport — place, inspect, and frame the dragon.
2. Properties Editor — import settings, materials, world, render, and output.
3. Shader Editor — simple node setup for material and environment.
4. Timeline — scrub frames, insert keys, and preview motion.
5. Outliner — keep scene names clean: Dragon, Camera, Sun, Path.

Recommended classroom habit

- Do one change at a time, then orbit / scrub / render-test.
- Rename objects before animation begins.
- Save a new version after successful import and again after the first still render.

If the interface feels large, reduce the mission: import correctly, look through the camera, press F12 for a still, then animate.

Step 1 — Import the provided dragon STL

Import

Classroom procedure

1. Open a clean Blender scene.
2. Choose File → Import → STL.
3. Select dragon.stl from the class folder.
4. Import once only; inspect before re-importing.
5. Rename the object to Dragon immediately.

Target result after this slide

- The mesh is visible in the viewport.
- The dragon is not unexpectedly rotated or microscopic.
- The Outliner contains a clearly named Dragon object.

*Import is a diagnostic step, not a beauty step.
Do not touch materials yet.*

Keyboard check: Numpad . on selected object to zoom to it.

Import settings that matter for STL

Import

Use these options intentionally, not by guesswork

Scale — start with 1.0; change only if the imported size is clearly wrong.

Scene Unit — use when scene units already matter to your workflow.

Facet Normals — keeps imported facet normals; useful when you intentionally want flat-looking faces.

Forward / Up Axis — use when the dragon arrives with wrong orientation. Blender expects Y forward and Z up.

Validate Mesh — helpful for suspicious files, but slower than normal import.

Rule of thumb for class: fix orientation first, then scale, and only then continue to shading or animation.

Before shading: prepare the imported mesh for the rest of the pipeline

Mesh prep

Preparation checklist

1. Check scale with a known reference or the grid.
2. Place the dragon near world origin for predictable camera and path work.
3. Make the dragon face its intended forward direction before path following.
4. Apply scale only after you are happy with size.
5. Save a new file version:
lecture9_dragon_scene.blend.

Why this matters later

- Wrong scale makes lighting and camera decisions harder to judge.
- Wrong forward axis produces awkward path-follow orientation.
- A messy origin or object name makes debugging much slower.

Do not animate a confusing object. Clean transforms first.

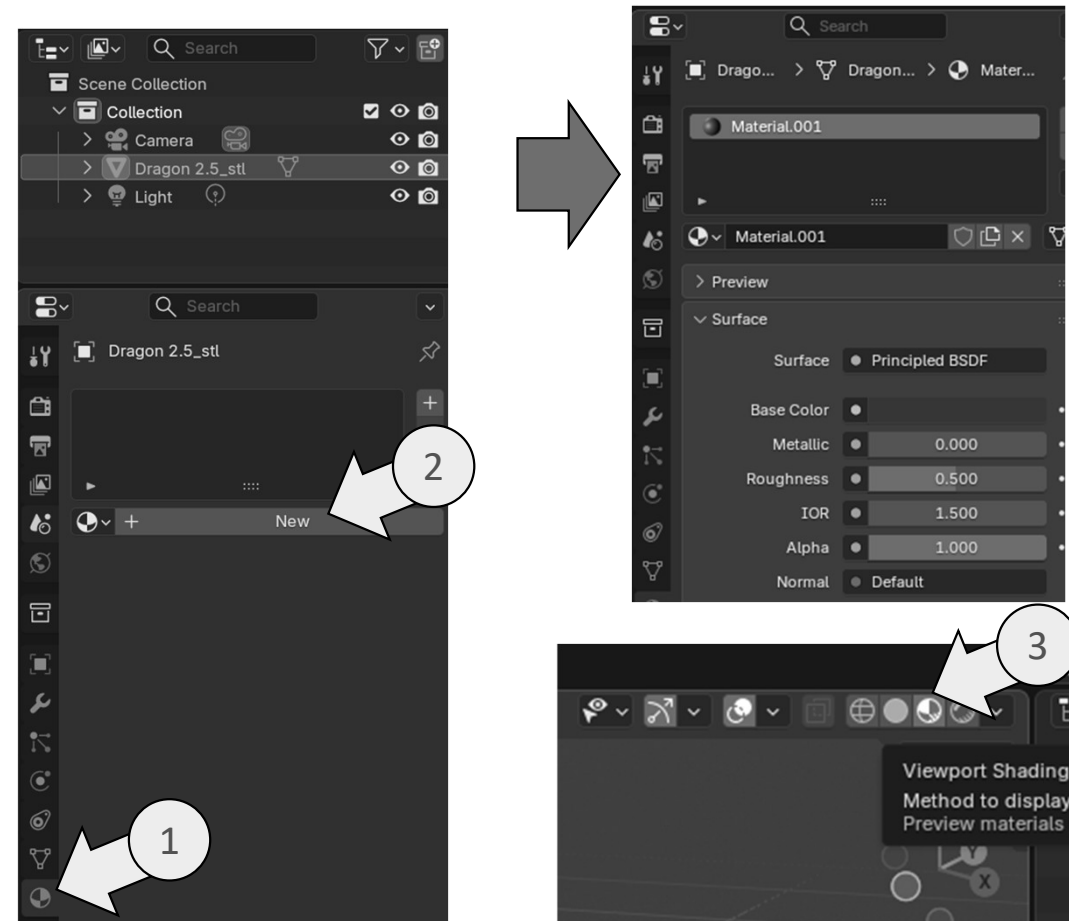
Step 2 — Build one reliable material with Principled BSDF

Material

Fast classroom recipe

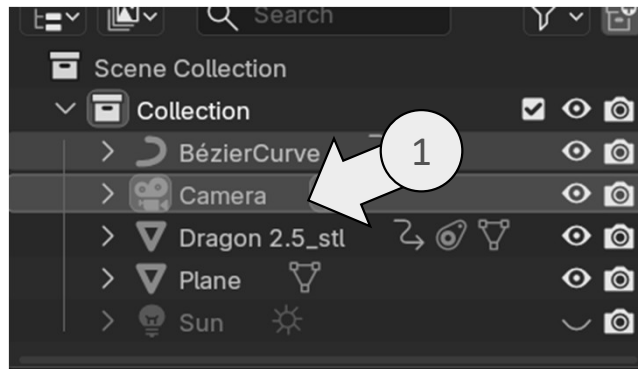
1. Create one material named Dragon_Mat.
2. Start with Base Color, Roughness, and Metallic only.
3. Keep values simple enough for quick visual comparison.
4. Do one preview render before adding complexity.

In this lecture, Principled BSDF is the “good default” because one node covers most common surface needs.

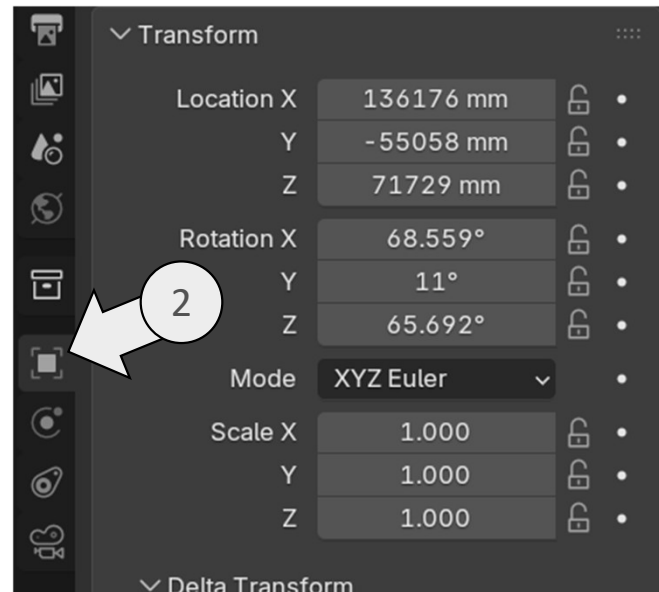


Setting camera and light

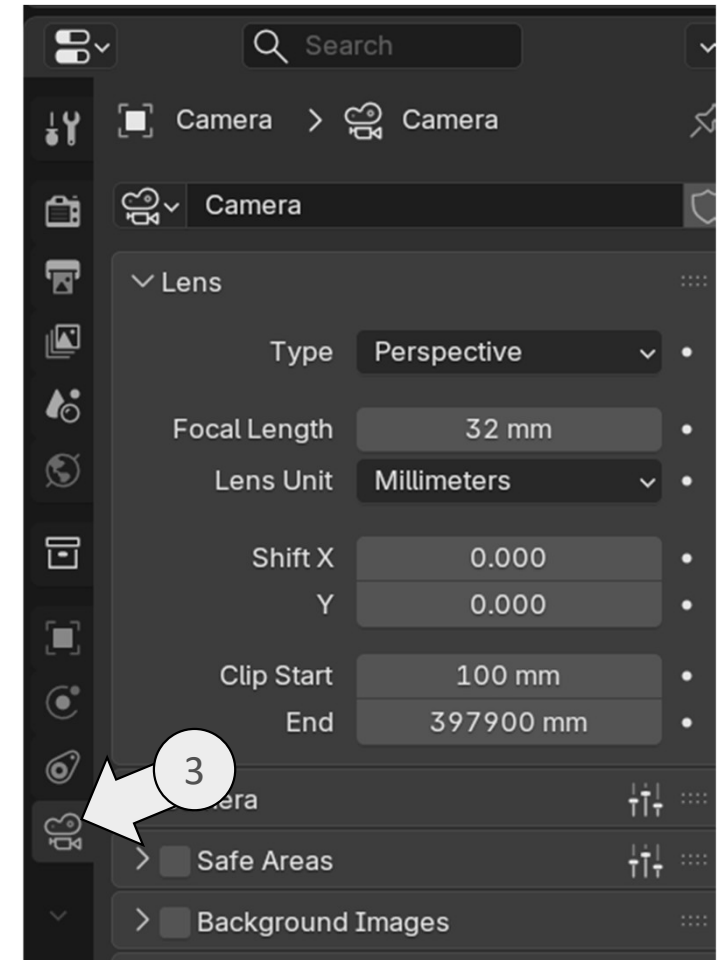
1) Select camera



2) Change direction of camera



3) Change clip start and end



Same for light, but this time we use light from HDR image

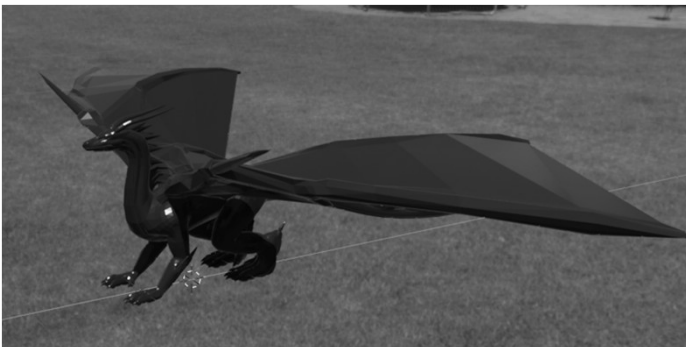
Step 3 — Render the object and its environment

Lighting

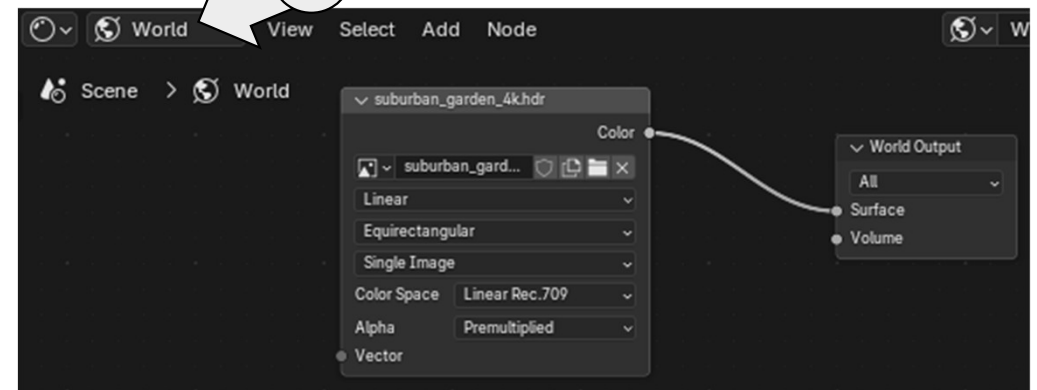
Scene target: the dragon must read clearly against a believable world setup.

- Use Material Preview or Rendered mode while adjusting.
- Set a World background first; add a light only if the form is still unclear.
- If an HDRI is available, connect it through an Environment Texture node.
- Make one fast still render now before animation begins.

Manual example: world lighting with HDR image



In **Shading** tab, world shader:
Delete existing background node
Right click add -> environment texture
Connect the node 1 world output



*Classroom shortcut: preview with quick settings;
reserve costly final quality for the end.*

Source of HDR file
<https://polyhaven.com/>

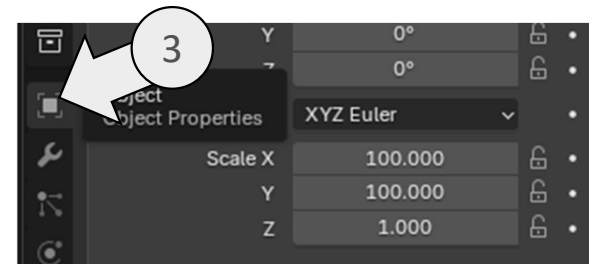
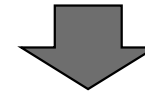
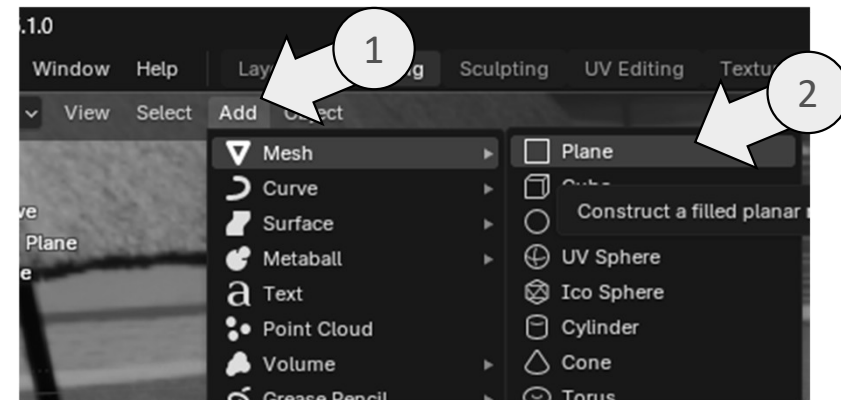
Step 3 — Render the object and its environment

Lighting

Create shadow using light source of HDR image

Add Plane,
Change size of plane in Object properties

In modeling tab -> add -> Mesh -> Plane



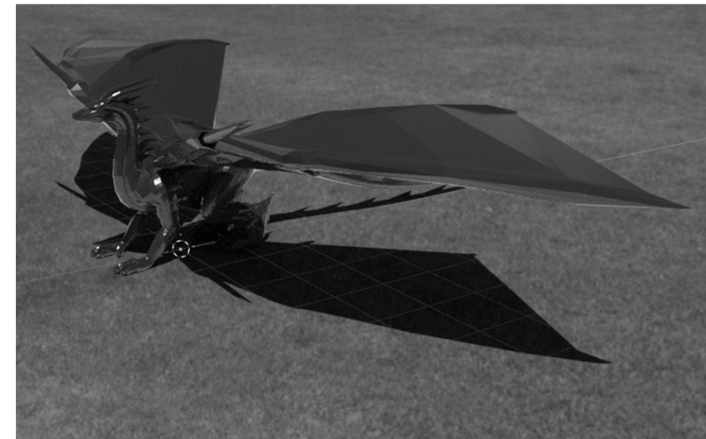
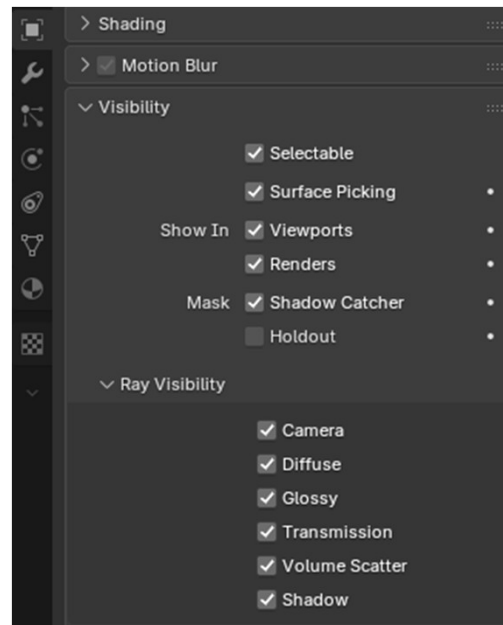
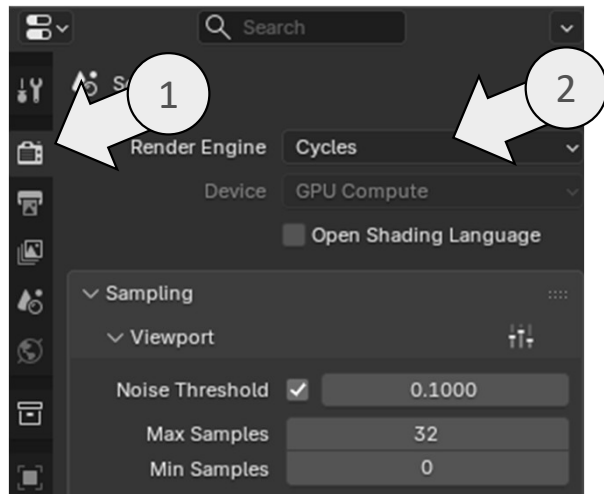
Step 3 — Render the object and its environment

Lighting

Create shadow using light source of HDR image

Use render engine: Cycles

In Plane object properties select “**Shadow catcher**”



Step 4 — Frame one successful still render before animating

Camera

Recommended classroom order

1. Place the camera.
2. Use a 3/4 view so silhouette and depth both read clearly.
3. Lock one “hero frame” before you think about motion.
4. Press F12 and evaluate shape, lighting, and background contrast.

Why a still render first?

- A bad still render becomes a bad animation many times over.
- Camera framing problems are easier to fix on one frame than across 100+ frames.
- This is the fastest checkpoint before the class invests in render time.

If frame 1 is not visually convincing, do not continue to animation yet.

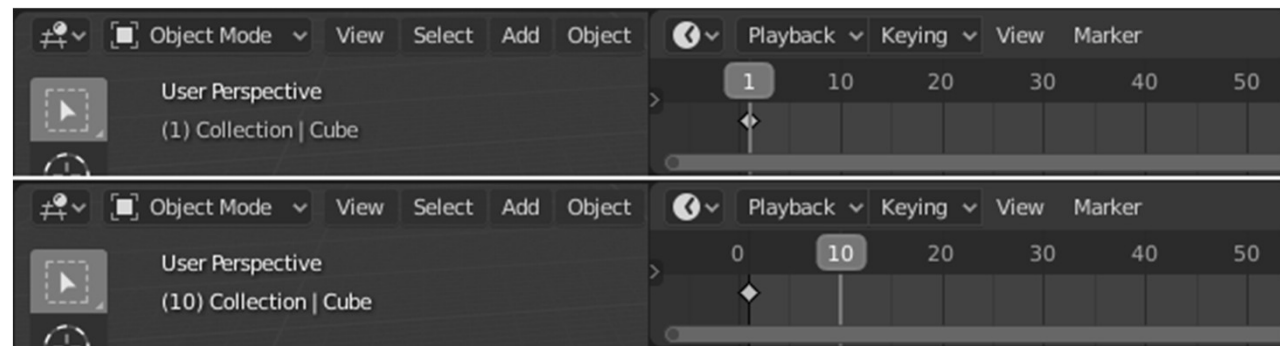
Animation plan (1/3) — start with the idea of keyframes

Animate

Minimum motion logic

1. Choose a start frame and an end frame.
2. Store one transform at the first frame.
3. Store a second transform later in time.
4. Let Blender interpolate the values between them.

In **Animation** tab

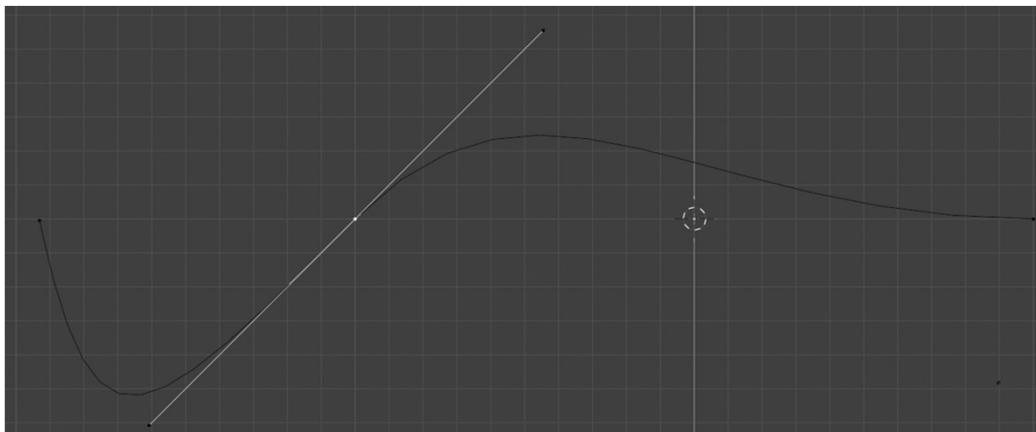


Drag keyframe pointer to desired frame and press “I” to create a new keyframes

Keyframes answer the question: “what value should this property have at this moment?”

Animation plan (2/3) — use a curve as the dragon's flight path

Animate

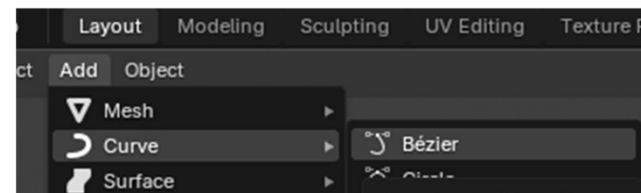


Official manual example: an object arranged to follow a curve

Why a curve is the right classroom tool here

- Students already know curve ideas from FreeCAD.
- The flight path becomes visible and editable in the viewport.
- The dragon motion reads as intentional rather than random translation.

Add Bezier curve



Click the node and press:

G to move the node

R to rotate tangent line

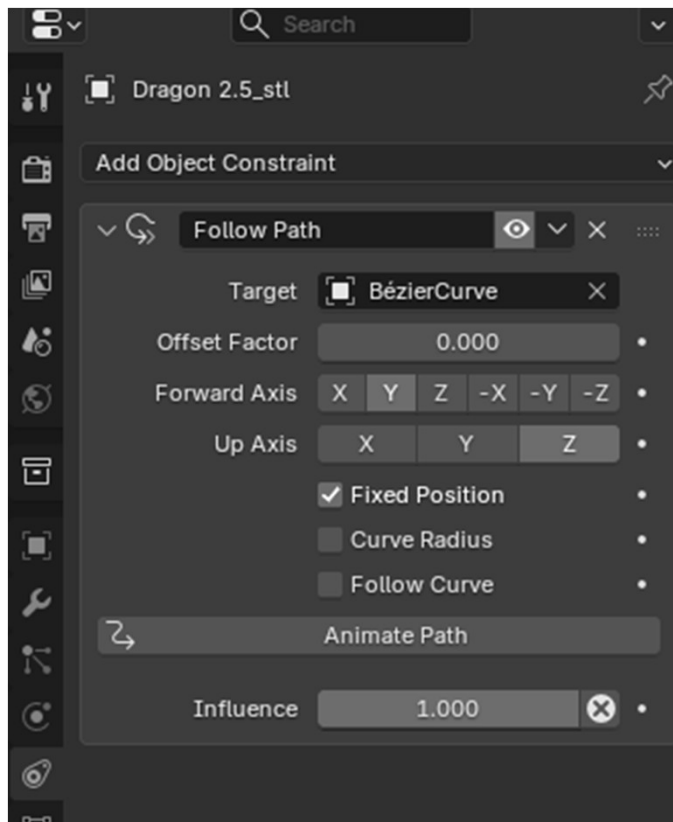
S to scale to tangent line

E to add point that extrudes from the selected point

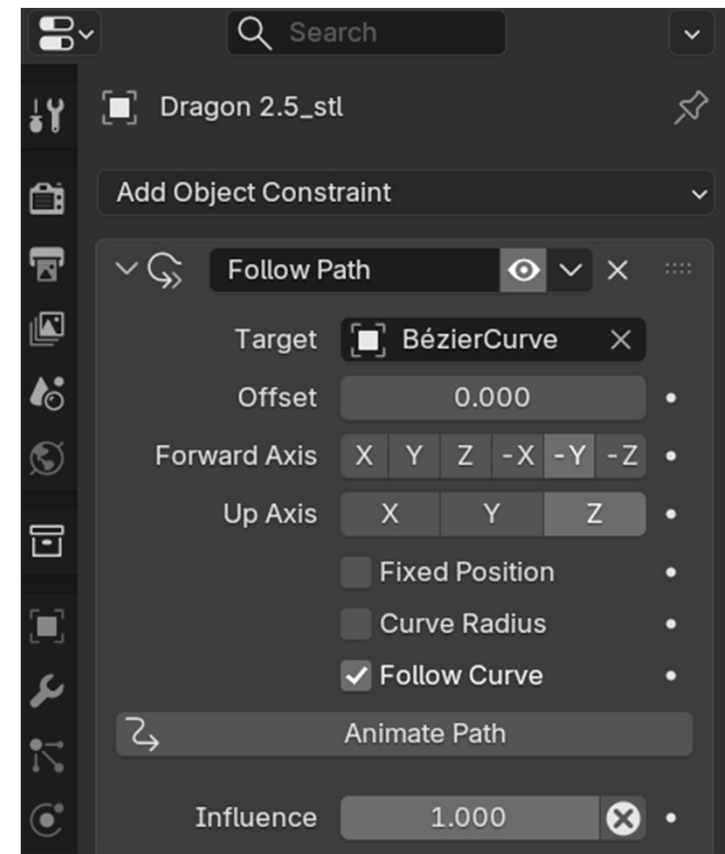
Animation plan (2/3) — use a curve as the dragon's flight path

Animate

Click dragon object -> constraint -> add object constraint
-> follow path -> choose BezierCurve



Select follow curve: forward axis -Y, up axis Z



Controlling speed and position on the path

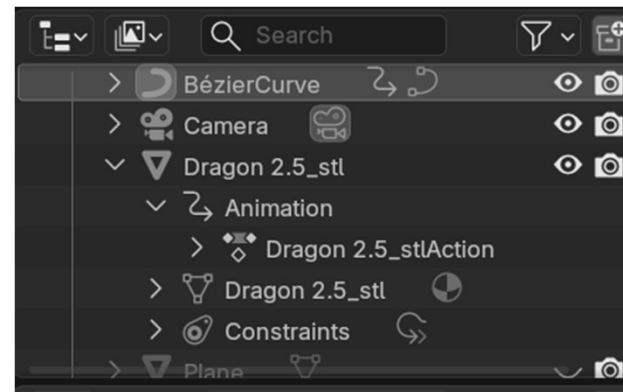
Animate

Operational rule for today

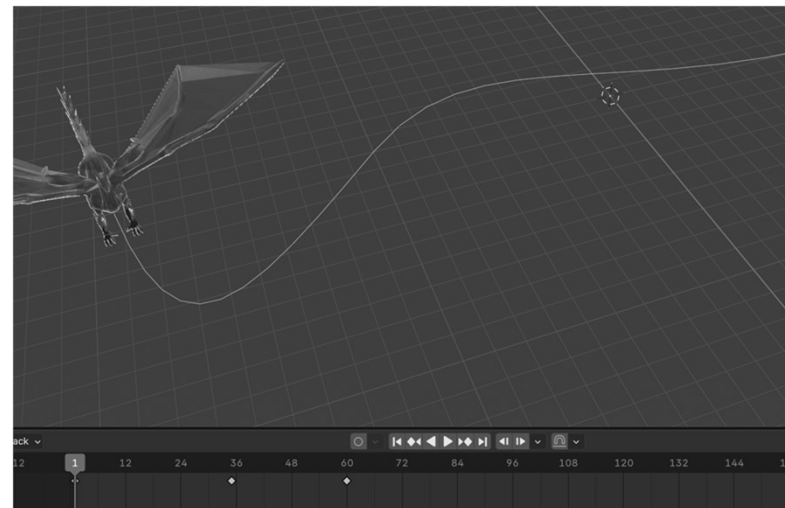
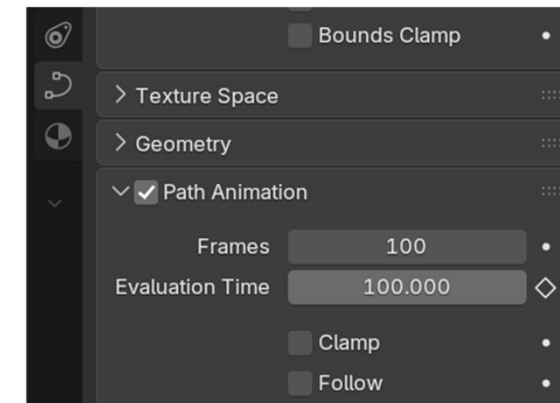
- Keep one simple frame range for the whole shot (Frames = 100).
- Use Evaluation Time (or path animation timing) to move from start to end.
- drag the frame pointer at each point and set corresponding evaluation time.

Path shape controls WHERE the dragon goes. Evaluation Time controls WHEN it reaches each point.

1) Select the curve

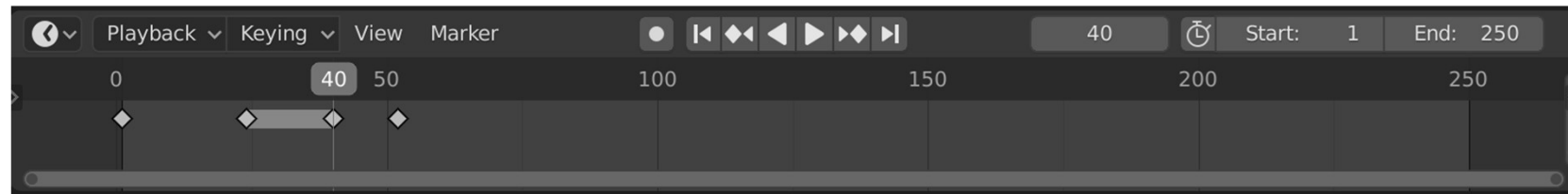


2) Select Data -> Path Animation panel



Timeline working rhythm for the classroom

Timeline



Manual timeline view

1. Set frame 1 and insert the first key (pressing I).
2. Jump to the end frame and insert the second key.
3. Scrub, play, stop, adjust, and repeat.
4. Keep playback focused on whether timing and readability are correct — not on final render quality.

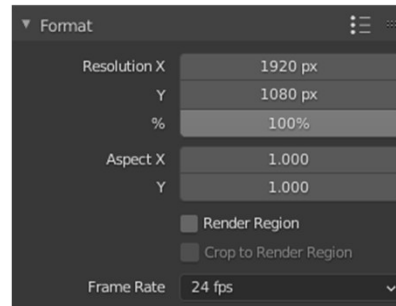
Step 5 — Set render size, frame rate, and frame range

Render setup

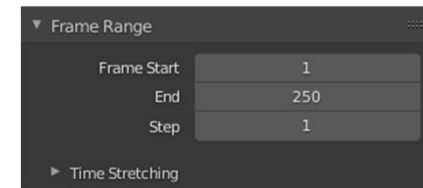
Classroom defaults to stabilize the pipeline

- Use a modest resolution for previews; scale up only after motion is correct.
- Pick one frame rate and keep it fixed across the shot.
- Define Start and End before final rendering so duration is explicit.

A short, clean animation beats a long, partially broken one.



Format



Frame Range

- Preview recommendation: keep output small enough for quick iteration.
- Check duration by $\text{frames} \div \text{fps}$ before pressing Render Animation.

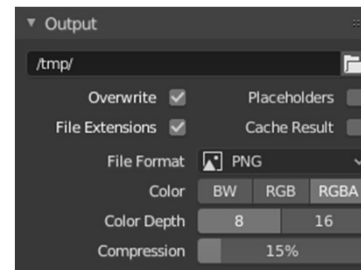
Output strategy — render frames first, encode video second

Render output

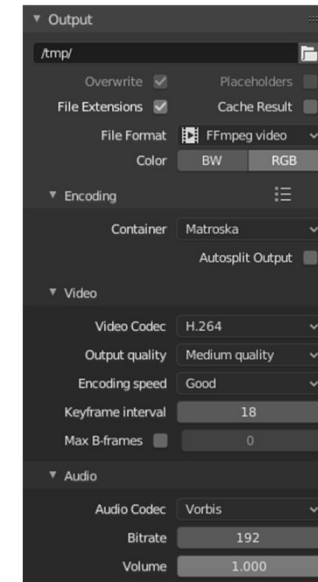
Recommended classroom workflow

1. Choose an output folder and clear naming pattern.
2. Render as image sequence first for safety and restartability.
3. After frames are correct, encode to a video container.

If rendering stops halfway, image sequence workflow lets you resume instead of losing the whole movie file.



Output panel



Encoding panel

Frames first = safer for long renders and easier to resume.
Video last = faster experimentation with codecs and file size.

Fast troubleshooting checklist during practice

Troubleshoot

Import

- Object not visible → frame selected.
- Wrong orientation → revisit import axis or object forward direction.
- Scale absurd → fix once, then apply scale.

Render

- Too dark → adjust world/light first.
- Flat look → check material roughness and light direction.
- Preview too slow → lower resolution or use faster engine for tests.

Animation

- Object off path → clean transforms / origin.
- Motion timing wrong → edit keys, not necessarily the path shape.
- Render too long → shorten frame range or preview first.

Instructor decision rule: debug one variable at a time — visibility, then orientation, then shading, then timing, then output.

In-class checkpoint and mini assessment

Checkpoint

Each student should show the instructor:

1. Imported Dragon object in viewport.
2. One still render that reads clearly.

Quick oral questions

- Why is frame sequence safer than direct-to-video?
- What is the difference between path shape and timing?
- Why do we verify one still render before full animation?

Wrap-up and suggested homework extension

Close

Today's pipeline in one sentence

Import the mesh cleanly, build one readable scene, animate motion with a path, and output frames safely.

Homework extension ideas

- Render a second camera angle of the same dragon flight.
- Try a different world lighting setup and compare mood.
- Refine timing by adjusting path speed instead of reshaping the path.
- If the course later covers rigs, compare object motion versus character deformation.

Remember the course connection

- Transforms and interpolation still matter.
- Lighting and material choices change perception.
- Rendering is the bridge from geometry to communication.